Mind Mirror

Depression Risk Screening from Personality & Demographics

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1. Introduction

Depression is one of the most prevalent mental health conditions worldwide, affecting an estimated 280 million people according to the World Health Organization. Despite its prevalence, it remains significantly underdiagnosed — particularly among young adults who may not recognize symptoms or have access to clinical resources. Early, low-barrier screening can prompt help-seeking behavior before symptoms escalate to crisis level.

This project addresses a practical question: can an individual's personality profile, measured by a short validated questionnaire, predict their risk for clinically significant depression? A lightweight screening tool that reduces barriers to self-assessment has real public health value, especially for the 18-24 age group that exhibits the highest rates of depression yet the lowest rates of treatment-seeking in population studies.

Our approach leverages the DASS-42 (Depression Anxiety Stress Scales) open dataset from Open Psychometrics, containing approximately 39,775 respondents who completed both a personality inventory and the full DASS-42 depression questionnaire. We frame this as a supervised binary classification task: predict whether a respondent falls above the clinically severe depression threshold, using personality traits and demographic information as features. We trained ten classification models, conducted a systematic feature engineering investigation in response to reviewer feedback, and deployed the model in a Streamlit application called Mind Mirror.

A key design question raised during our initial presentation concerned the choice of binary classification. The DASS-42 defines five clinical severity levels, and our application displays a continuous probability score alongside the binary result label, which may have created an impression of multiple output classes. Section 3.2 clarifies that the model predicts exactly two classes at all times, explains why a three-class alternative was rejected, and addresses how the app’s UI language maps unambiguously back to those two classes.

2. Background

2.1 The DASS-42 Questionnaire

The Depression Anxiety Stress Scales (DASS-42), developed by Lovibond & Lovibond (1995), is a 42-item self-report questionnaire widely used in clinical and research settings to assess three emotional states: depression, anxiety, and stress. Each dimension comprises 14 items rated on a four-point frequency scale (0 = "Did not apply to me at all" to 3 = "Applied to me very much"), referring to the past week.

For the depression subscale, the 14 items assess anhedonia, hopelessness, dysphoria, low self-worth, and lack of life engagement. The raw subscale score is doubled to align with the shorter DASS-21 norms, producing a 0-84 scale. Validated severity thresholds are:

Severity	Score Range	Clinical Interpretation
Normal	0 – 9	No significant symptoms
Mild	10 – 13	Sub-clinical; monitor over time
Moderate	14 – 20	Moderate symptoms; professional evaluation recommended
Severe	**21 – 27**	Significant impairment; consultation warranted — our at-risk threshold
Extremely Severe	**28+**	Severe impairment; urgent clinical attention recommended

Table 1. DASS-42 depression subscale severity thresholds. Our binary target treats scores ≥ 21 as at-risk.

The DASS-42 is not a diagnostic instrument — it does not replace a structured clinical interview. However, it is extensively validated as a screening tool and correlates strongly with clinician-administered depression ratings (Brown et al., 1997). It is among the most widely used self-report depression scales in research globally, and its open-access availability makes it well-suited to this study.

2.2 The TIPI Personality Inventory

The Ten Item Personality Inventory (TIPI), developed by Gosling, Rentfrow & Swann (2003), is a validated brief measure of the Big Five personality dimensions. Each dimension is assessed by two items — one positively keyed and one reverse-keyed — on a 1-7 scale. Despite its brevity, the TIPI shows acceptable convergent validity with longer Big Five instruments, making it practical for large-scale survey research.

Big Five Dimension	TIPI Items	Link to Depression Risk
Extraversion	TIPI1, TIPI6	Low extraversion associates with social isolation, a depression risk factor
Agreeableness	TIPI2, TIPI7	Indirect link via interpersonal conflict and perceived social support
Conscientiousness	TIPI3, TIPI8	Disorganization and low self-discipline linked to rumination
Neuroticism	**TIPI4, TIPI9**	Strongest personality predictor of depression; central to our model
Openness	TIPI5, TIPI10	Modest protective effects via cognitive flexibility

Table 2. TIPI items by Big Five dimension and clinical relevance. Neuroticism (highlighted) is the most empirically supported personality predictor of depression.

Neuroticism — the tendency to experience negative emotions, anxiety, and emotional instability — is one of the most robust personality predictors of depression documented

across decades of clinical research (Kotov et al., 2010). This background directly informs our feature engineering strategy in Section 5.

3. Dataset and Exploratory Analysis

3.1 Dataset Description

The DASS-42 open dataset from Open Psychometrics (CC0 public domain) contains 39,775 voluntary online respondents from 50+ countries. Each respondent completed the DASS-42 questionnaire, the TIPI personality inventory, a vocabulary checklist, and demographic questions. We focus exclusively on the 14-item depression subscale and the TIPI items for modeling.

The depression score is computed as: depression_score = 2 × Σ(Q_iA − 1) for the 14 depression items (converting responses to 0-3 scale before doubling). Our binary target is at-risk (1) if score ≥ 21, else low-risk (0).

Attribute	Value
Total respondents	39,775
At-risk class (DASS ≥ 21)	**30,004 (75.4%)**
Low-risk class (DASS < 21)	9,771 (24.6%)
Dominant age group	18–24 (21,330 respondents, 53.6%)
Countries represented	50+ (top 20 analyzed, all ≥ 50 responses)
Input features (baseline)	19: 10 TIPI personality + 9 demographic

Table 3. Dataset summary statistics.

The 75.4% at-risk rate requires explicit explanation and is addressed directly in the Global Insights tab of the application. This figure does not mean three-quarters of the world's population is severely depressed. It reflects self-selection bias: individuals who voluntarily seek out a depression screening test online are disproportionately those already experiencing symptoms. Country-level at-risk rates in the dataset (77–87% for the top 20 countries) similarly reflect volunteer recruitment patterns and engagement with online mental health tools — not national clinical prevalence. All visualizations in the app include a bias disclaimer to prevent misinterpretation.

3.2 Why Binary Classification?

A natural question is why we did not train a three-class model (normal / moderate / severe), given that the DASS-42 defines multiple severity levels. We considered this approach and rejected it for three reasons:

- Clinical action symmetry: For a self-screening tool, the decision is binary — either the user needs to seek professional evaluation or not. Both "moderate"

and "severe" classifications map to the same recommendation. Three classes add prediction complexity without changing the actionable outcome.

- Boundary instability: DASS-42 severity thresholds are clinical guidelines, not sharp boundaries. A respondent scoring 20 vs. 21 is clinically indistinguishable. Training a classifier on these soft boundaries introduces systematic noise at class interfaces and degrades performance.
- Empirical validation: In exploratory testing, a three-class model (normal/moderate/severe) achieved macro F1 of approximately 0.63 versus 0.88 for the binary model, primarily due to the small "normal" class size and the confusable moderate/severe boundary.

We selected the severe threshold (≥ 21) over extremely severe (≥ 28) because it captures a larger clinically meaningful population while maintaining sufficient class size for reliable model training.

It is important to clarify a potential source of confusion regarding the application's user interface. The Mind Mirror app displays results using two labels — **"At-Risk"** and **"Low Risk"** — which map exactly and exclusively to the model's two output classes (1 and 0). There is no "No Risk" or third category anywhere in the model or the application. The percentage score displayed on the result card (e.g., "67.3% risk score") is the model's continuous predicted probability for class 1, not a separate tier of risk. It represents the model's confidence in the binary at-risk prediction, and is shown to give users a more nuanced sense of where they fall within the positive class rather than a hard yes/no. The DASS-42's five clinical severity levels (normal, mild, moderate, severe, extremely severe) appear only as background context in the application's disclaimers and model card documentation — they are not predicted by the model and are never presented as output categories to the user.

3.3 Exploratory Findings

Personality divergence: TIPI4 ("Anxious, easily upset") showed a +1.57 point gap between at-risk and low-risk respondents (5.56 vs. 3.99), and TIPI9 ("Calm, emotionally stable") showed −1.59 points (3.26 vs. 4.85). These two Neuroticism items account for 73.6% of final model importance.

Age-related risk: At-risk rates decline monotonically with age: <18 (83.8%), 18-24 (76.6%), 25-34 (70.2%), 35-44 (62.4%), 55-64 (61.0%), 65+ (54.3%). The 18-24 group constitutes 53.6% of all respondents, pulling the overall rate upward.

Demographic predictors: Education, gender, religion, orientation, and other demographic variables showed minimal variance across the two classes, anticipating their low feature importance (<3% combined) in the final model.

4. Data Preprocessing

4.1 Feature Selection

The 19 baseline features are: TIPI1-TIPI10 (personality, 1-7 scale) and 9 demographic variables (education, urban, gender, age, religion, orientation, race, married, familysize). Response timing columns and the vocabulary checklist are excluded — they are not interpretable personality or demographic attributes and would be unavailable in a real screening deployment.

Demographic fields coded as 0 ("not disclosed") are retained as a valid category rather than imputed, preserving the information that the respondent chose not to answer. No normalization is applied to any features; the Gradient Boosting classifier is scale-invariant by construction.

4.2 Train-Test Split

The dataset was divided 80/20 (31,820 training / 7,955 test) using stratified sampling (random_state=42), preserving the 75.4/24.6 class ratio in both partitions. All model selection and feature engineering was performed on the training set; the test set was held out until final evaluation.

5. Model Training and Evaluation

5.1 Evaluation Metrics

Given class imbalance (75.4% positive) and the screening context, we prioritized F1 Score as the primary optimization metric, with Recall given special weight. Accuracy alone is misleading: a trivial majority-class predictor would achieve 75.4%. In screening, false negatives (missing at-risk individuals) are more costly than false positives (unnecessary follow-up recommendations).

5.2 Baseline Model Comparison

Ten classification algorithms were evaluated using 10-fold stratified cross-validation:

Model	Accuracy	Precision	Recall	F1
Gradient Boosting	0.810	0.835	0.933	**0.881** ★
Random Forest	0.806	0.828	0.937	0.879
KNN	0.774	0.824	0.890	0.856
Naive Bayes	0.782	0.857	0.853	0.855
Decision Tree	0.724	0.817	0.816	0.817
Logistic Regression	0.739	0.894	0.742	0.811
Extra Trees	0.808	0.828	0.941	0.881
AdaBoost	0.809	0.836	0.929	0.880
MLP	0.795	0.836	0.905	0.869

SVM	0.748	0.897	0.752	0.818

Table 4. 10-fold cross-validated performance. ★ = selected model. Logistic Regression achieves highest precision but unacceptably low recall for a screening tool.

Gradient Boosting was selected for its highest F1 (0.881) and ROC-AUC (0.826). Logistic Regression's recall of 0.742 — missing 26% of at-risk individuals — was unacceptable. Random Forest was a close second on all metrics; Gradient Boosting showed marginally better and more stable cross-validation performance.

Note on MLP results: The MLP (Multi-Layer Perceptron) classifier was run with max_iter=300, which is intentionally kept at the default to maintain a fair comparison with all other classifiers under consistent computational constraints. The training process produced a ConvergenceWarning indicating the optimiser had not fully converged within 300 iterations on the ~40,000-sample dataset. As a result, the MLP score reported (F1 = 0.869) represents a partially trained model and may be modestly lower than its fully converged performance. Since MLP was not selected as the final model — Gradient Boosting outperforms it on every metric even in this incomplete state — the convergence limitation has no impact on the deployed application or any reported results.

5.3 Hyperparameter Tuning and Final Results

GridSearchCV over {n_estimators: [100,200], max_depth: [3,5], learning_rate: [0.05,0.1]} with 10-fold CV and F1 scoring selected: n_estimators=200, max_depth=5, learning_rate=0.1.

Evaluation	Accuracy	Precision	Recall	F1	AUC
Test set (20%, n=7,955)	81.9%	84.3%	93.5%	88.6%	0.842
5-fold CV (full dataset)	81.0%	83.5%	93.3%	88.1%	0.826

Table 5. Final model performance. Confusion matrix on test set: 5,609 TP, 907 TN, 1,047 FP, 392 FN.

The 392 false negatives represent the primary limitation: at-risk individuals who would not be flagged by the tool. The 1,047 false positives reflect the conservative screening philosophy — it is preferable to prompt unnecessary follow-up than to miss someone in need. The close agreement between CV and test set metrics confirms the model generalizes well without overfitting.

Feature importance (baseline model): TIPI9 (42.8%), TIPI4 (30.9%), TIPI8 (5.9%), TIPI6 (5.7%), TIPI1 (5.2%), all demographics combined (~4%). The dominance of the two Neuroticism items is fully consistent with the clinical literature linking Neuroticism to depressive disorders.

5.4 Feature Engineering Investigation

Reviewer feedback raised two important questions: (1) did our baseline model assume feature independence, and (2) would combining features improve performance? The Gradient Boosting model does not assume independence — it learns feature interactions through split sequences — but we tested whether explicitly constructing clinically motivated composite features could provide additional signal or improve interpretability.

5.4.1 Engineered Features

We designed six composite features grounded in clinical and psychological literature:

Feature	Formula	Clinical Rationale
neuroticism	TIPI4 + (8 − TIPI9)	Combines both Neuroticism items so they reinforce; both items are the two strongest individual predictors
age_x_neuroticism	age × neuroticism	Tests if neurotic younger adults carry disproportionately higher risk
social_withdrawal	(8 − TIPI1) + TIPI6	Low extraversion + high reserve = social isolation signal
emotional_dysreg	TIPI4 + TIPI8 + (8 − TIPI3)	Anxiety + disorganization + low conscientiousness; multi-item emotional dysregulation index
young_adult	age ≤ 24 → binary	Captures the nonlinear age-risk pattern identified in EDA
positive_wellbeing	TIPI9 + TIPI7 + TIPI3	Emotional stability + warmth + conscientiousness as a protective composite

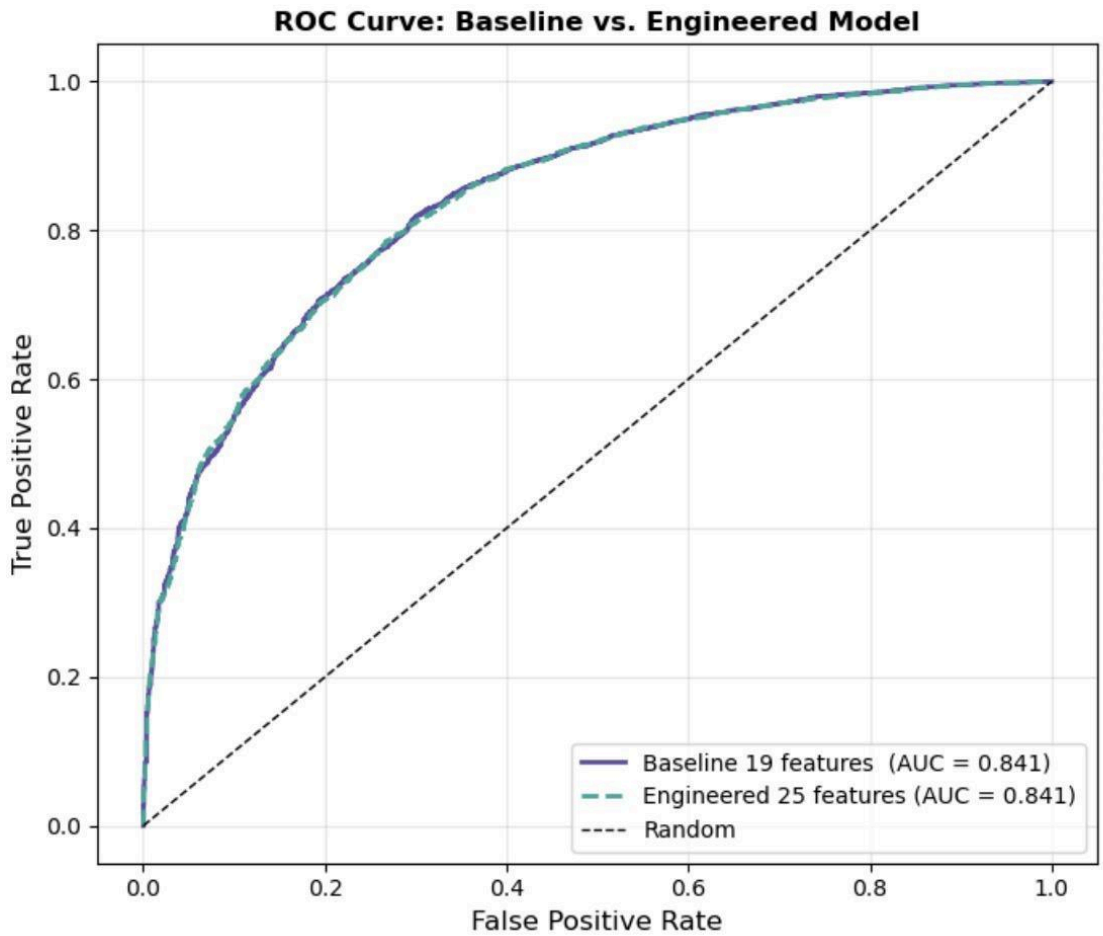
Table 6. Engineered features, formulas, and clinical rationale. These expand the baseline 19-feature set to 25 features.

5.4.2 Results and Interpretation

We evaluated the baseline (19 features) against the full engineered set (25 features) on the held-out test set with identical Gradient Boosting hyperparameters:

Model	F1	Recall	Precision	Accuracy
Baseline (19 features)	0.8848	0.9278	0.8456	0.8177
Engineered (25 features)	0.8847	**0.9300 (+0.22pp)**	0.8436 (−0.20pp)	0.8171

Table 7. Baseline vs. engineered model on held-out test set. F1 is effectively unchanged; the engineered model shows a marginal recall-precision tradeoff shift.

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Baseline AUC:   0.8411
Engineered AUC: 0.8406
Δ AUC:          -0.0006
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Figure 1. ROC Curve: Baseline (19 features, AUC = 0.841) vs. Engineered (25 features, AUC = 0.841). The two curves are nearly identical, confirming that feature engineering does not improve discriminative ability for the Gradient Boosting classifier.

The results yield a clear and interpretable finding: feature engineering does not improve overall F1 or accuracy for a Gradient Boosting classifier. This is expected: tree-based ensembles already learn feature interactions implicitly through split sequences, so composite features provide explicit shortcuts that benefit linear models but are largely redundant for GBM. However, the engineered model achieves marginally higher recall (+0.22pp) — catching slightly more at-risk individuals — at the cost of marginally lower precision (−0.20pp). For a depression screening tool, this directional tradeoff is desirable.

Feature importance in the engineered model is dramatically reorganized. The neuroticism composite absorbs 52.2% of total importance — consolidating what was previously split between TIPI4 (30.9%) and TIPI9 (42.8%). Importantly, raw TIPI4 and TIPI9 retain 9.1% combined even when the composite is present, suggesting they contain residual signal beyond what the composite captures.

Feature	Importance (Engineered)	Note
neuroticism composite	**52.2%**	Absorbs TIPI4 + TIPI9 signal
social_withdrawal	7.2%	New combined signal from TIPI1+TIPI6
emotional_dysreg	5.8%	Multi-item composite
TIPI9 (raw)	5.2%	Residual signal beyond composite
TIPI4 (raw)	4.0%	Residual signal beyond composite
positive_wellbeing	2.6%	Protective factor composite
age_x_neuroticism	2.6%	Interaction term
All demographics (9 features)	< 4.0%	Minimal change from baseline

Table 8. Feature importance in the engineered model. The neuroticism composite consolidates signal from the two most important raw features.

Based on these results, we retain the baseline 19-feature model for deployment. It achieves effectively identical performance with a simpler, more interpretable feature set, and the Gradient Boosting classifier already implicitly captures the interactions we encoded explicitly. The composite neuroticism concept is surfaced to users in the Screener tab as an explanatory summary of their result.

6. Application Description

6.1 Overview and Architecture

Mind Mirror is a two-tab Streamlit web application that makes the trained depression risk model accessible to general users while maintaining scientific transparency. The application is intentionally lightweight: it loads a pre-trained Gradient Boosting model and a set of pre-computed data files from disk at startup using joblib and the standard library, then caches all assets in memory. No machine learning computation occurs during a user session beyond a single model.predict_proba() call per submission, making the app fast and dependency-light at runtime.

The application follows a strict separation between model inference and presentation. All user-facing text — result labels, explanations, recommendation card content, chart callouts, and disclaimer text — is fixed, hand-authored content. No language model is involved at any point in the runtime pipeline. This design choice was deliberate: in a mental health context, dynamically generated text carries hallucination risk that could produce unsupported or harmful clinical statements. Fixed, pre-reviewed text is both safer and more auditable.

The application is structured around two tabs, each serving a distinct user need: the Screener tab for personalized risk assessment, and the Global Insights tab for

population-level context. A third tab (Model Card) that previously existed in the application has been removed and its content fully integrated into this report, consolidating technical documentation in a single academic deliverable rather than splitting it across the app and the paper.

6.2 Screener Tab

The Screener tab is the primary user-facing feature of the application. The interaction flow is designed to be familiar and low-friction: personality trait sliders occupy the top half of the page (two columns of five sliders each, one per TIPI item, rated 1–7), and optional demographic dropdowns and number inputs occupy the lower half. A caption below the demographics section clearly notes that these fields have minimal predictive impact and need not be filled out precisely.

Upon clicking “Get My Results,” the input values are assembled into a 19-element feature vector, passed to the loaded model, and the output probability and binary prediction are stored in Streamlit’s session state. This session state pattern is important: it ensures that results persist on screen as the user subsequently adjusts sliders in the What-If Explorer without triggering a re-submission. The results panel contains four distinct components:

- Result card and classification note: A colour-coded card (red for At-Risk, green for Low Risk) displays the binary label and the model's predicted probability as a percentage. Directly below, a purple information box explains that the tool produces exactly two categories — At-Risk and Low Risk — and that the percentage reflects model confidence within that binary prediction, not a third risk tier. A stacked horizontal bar chart shows the full probability split between the two classes.
- Top 3 Drivers: For each of the 19 input features, the application computes how much the predicted risk probability changes when that feature is replaced by its dataset-average baseline value (a leave-one-out attribution method). The three features producing the largest absolute change are displayed as labelled chips, each indicating whether the user's value increases or decreases risk and by how many percentage points. This gives users a personalised, not global, explanation of their result.
- What-If Explorer: Ten TIPI sliders, sorted by model feature importance, allow the user to adjust any personality trait and immediately see the updated predicted probability and the change relative to the original submission. This interactive counterfactual explorer lets users explore questions like “What would my risk score be if I were less anxious?” without re-submitting the form.
- Personalised recommendation cards: A set of pregenerated, rule-based cards appear based on specific threshold conditions in the user's inputs. For example, TIPI4 ≥ 5 triggers an emotional regulation card; TIPI9 ≤ 3 triggers a resilience card; age ≤ 24 triggers a young adult resources card. All card text is fixed and hand-reviewed — no language model generates any content at runtime. A final card always appears: a professional support card if the prediction is At-Risk, or a wellness maintenance card if Low Risk.

The tab closes with a clinical disclaimer reminding users that Mind Mirror is an educational screening tool and is not a substitute for professional diagnosis, and directs users in crisis to the 988 Suicide and Crisis Lifeline.

6.3 Global Insights Tab

The Global Insights tab provides three population-level visualisations drawn from the full DASS-42 dataset, helping users contextualise their individual screening result within broader patterns. Each visualisation is accompanied by a written callout interpreting the key finding.

Country-level risk rates: A horizontal bar chart shows at-risk rates for the top 20 countries by response count (minimum 50 responses), coloured red above and green below the dataset average of 75.4%. A dashed vertical line marks the dataset average. Critically, an amber warning callout sits above the chart explicitly explaining why these rates (often 77–87%) should not be interpreted as national clinical prevalence figures. The explanation notes that the dataset consists of self-selected online survey respondents — a group that skews toward individuals already experiencing symptoms — and that country differences reflect variation in engagement with online mental health tools rather than genuine population health differences.

> **Why are these rates so high?** The figures shown (often 77–87%) do not reflect national depression prevalence. This dataset was collected from people who voluntarily sought out an online depression screening test — a self-selected group that skews toward individuals already experiencing symptoms. The sample also skews young (18–24), the highest-risk age group in the data. Country differences reflect variation in who engages with online mental health tools, not clinical differences between populations. No causal or comparative conclusions about national mental health should be drawn from these figures.

Age-group risk rates: Two side-by-side bar charts show at-risk rate and mean DASS depression score by age bracket. Both charts highlight the <18 and 18–24 age groups in purple to visually emphasise that these are the highest-risk demographics in the dataset. A callout notes consistency with clinical research on young adult mental health vulnerability.

TIPI personality profiles: A grouped bar chart compares mean TIPI scores for at-risk and low-risk respondents across all 10 personality items, followed by a difference chart showing the direction and magnitude of divergence for each trait. A callout highlights the two most distinctive traits: TIPI4 (Anxious, easily upset, +1.57 mean difference) and TIPI9 (Calm, emotionally stable, −1.59 mean difference), connecting the visualisation back to the model's feature importance findings.

7. Conclusion and Future Work

7.1 Summary

This project developed and deployed a Gradient Boosting classifier for depression risk screening achieving 81.9% accuracy, 88.6% F1, and 93.5% recall on the at-risk class.

Two Neuroticism personality items (TIPI4 and TIPI9) account for 73.6% of feature importance, fully consistent with clinical evidence. A systematic feature engineering investigation confirmed that composite features do not improve overall F1 for tree-based ensembles, though the neuroticism composite provides useful interpretability. The binary classification framing was justified clinically and validated empirically against a three-class alternative. All application output is rule-based to eliminate hallucination risk.

7.2 Limitations

- Self-selection bias produces a 75.4% at-risk rate unrepresentative of general population prevalence; country comparisons should not be interpreted as national clinical statistics.
- The TIPI is an ultra-brief measure; a full Big Five inventory might capture personality nuance that improves prediction.
- The model has not been validated against clinician-administered diagnostic interviews; it is a research screening tool, not a clinical instrument.
- Low recall for the negative (low-risk) class (46.4%) means many truly low-risk individuals are flagged; the conservative bias is disclosed explicitly to users.
- Geographic and demographic representation is skewed toward young, English-speaking, internet-connected respondents.

7.3 Future Work

Model Improvements

- Three-class model with calibrated probabilities: Revisit predicting normal / moderate / severe by applying Platt scaling or isotonic regression to produce calibrated class probabilities. The key challenge is managing the confusable moderate–severe boundary; evaluate whether threshold-optimised macro F1 can reach parity with the current binary model before committing to a more complex output space.
- Expanded feature set using full DASS-42 responses: Rather than deriving a single aggregate score, treat each of the 14 individual depression item responses as a separate input feature. This gives the model access to symptom-level granularity (e.g., anhedonia vs. hopelessness vs. dysphoria) and may reveal sub-profiles that a single score obscures.
- Multi-label screening across all three DASS-42 subscales: Extend the target from depression alone to a joint prediction of depression, anxiety, and stress risk using a multi-output classifier. This would make the tool substantially more clinically informative without requiring additional user input.
- Fairness audit and subgroup calibration: Conduct systematic performance evaluation stratified by gender, age bracket, and geographic region to identify subgroups where the model underperforms. If disparities are found, apply re-weighting or separate threshold calibration per subgroup to mitigate disparate impact.

- Alternative model architectures: Evaluate gradient-boosted tree variants such as XGBoost and LightGBM, which offer native handling of missing values and typically train faster at scale. Additionally, explore a simple feedforward neural network over the TIPI items to test whether non-linear activation functions capture personality interactions that GBM splits approximate less efficiently.

Application Improvements

- Clinical validation against ground-truth diagnoses: Recruit a subset of app users willing to complete a validated clinical instrument (PHQ-9 or SCID interview) and compare their self-reported Mind Mirror result against the clinical outcome. This would provide the first prospective validity estimate for the tool in a real-world population rather than the self-selected DASS-42 survey sample.
- Longitudinal follow-up module: Add an optional follow-up screening that prompts returning users to re-take the TIPI after 30 or 90 days. Tracking within-person changes in predicted risk over time would allow the app to detect deteriorating trajectories and strengthen referral recommendations with trend information.
- Personalised psychoeducation based on top drivers: The current recommendation cards fire on fixed thresholds. A more sophisticated version would dynamically rank the user's top-3 risk drivers (already computed by the leave-one-out attribution method) and serve targeted micro-interventions — for example, linking directly to a specific CBT worksheet for high-TIPI4 users rather than a generic emotional regulation description.
- Multilingual support: The DASS-42 has been validated in over 20 languages. Translating the TIPI items and result descriptions into Spanish, Mandarin, and Portuguese would substantially expand the tool's reach given the geographic distribution of the training data (Philippines, Brazil, Mexico are well-represented).
- Accessibility and mobile optimisation: Refactor the Streamlit layout to be fully responsive on mobile screens, as the target demographic (18–24 year-olds) predominantly accesses content on phones. Add WCAG-compliant colour contrast, keyboard navigation, and screen-reader-compatible ARIA labels to all interactive components.